Thesis Update #2

Milan Chhetri

Mohammad Ghamari

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Research Paper #1 :
“Non-invasive blood glucose monitoring using PPG signals with various deep learning models and implementation using TinyML”[1]

Non-invasive Glucose Monitoring

Overview:

- Extracted 1-second windows around systolic and diastolic peaks
- Train and compare Conv1D models
  - ResNet34 (best)
  - VGG16
  - CNN-LSTM-Attention
- Deployed on STM32H743IIT6
  - Low-power
  - 7 MB size
  - QSPI External Flash
  - 6.4 second inference

Non-invasive Glucose Monitoring

Methods:

- Datasets
  - VitalDB: 6388 patients - 70% training, 15% validation, 15% testing
  - MUST: 67 raw PPG signals with invasive BGL labels – External test dataset
- Preprocessing:
  - Downsample PPG from 500 Hz to 100 Hz
  - Synchronize PPG with BGL measurements
  - Crop 8 min before/after BGL measurement
  - Remove invalid segments
  - Forward Fill missing data
  - Butterworth bandpass filtering from 0.5 Hz to 8 Hz

Non-invasive Glucose Monitoring

Methods:

- Segmentation:
  - Extract 1-second windows around systolic and diastolic peaks
  - Perform cosine-similarity filtering
  - Preserve clean, useful information on PPG structure
- Models:
  - Modified ResNet34 to perform Conv1D
  - Residual blocks mitigate vanishing gradients
    - $Block\ Output = F(x) + x$
    - $x = previous\ block\ output\ , F(x) = processed\ output$
  - Skip connection helps gradient flow backward

Non-invasive Glucose Monitoring

Results:

- Clarke Error Grid Analysis – VitalDB test dataset
  - ResNet34 10-sec window:
    - 68.0% Zone A, 31.4% Zone B, RMSE = 28.4 mg/dL
  - ResNet34 1-sec window
    - 72.6% Zone A, 25.9% Zone B, RMSE = 27.48 mg/dL
- 1-sec window better, but nonzero Zone C/D/E errors
- Embedded Deployment
  - Post-Quantized Model exceeded internal flash →external QSPI flash
  - 6.4 second inference
  - 7 MB model size post-optimization

Non-invasive Glucose Monitoring

Takeaways:

- Subject dependent
  - PPG signals vary with physiology and tissue composition
  - Model may learn subject-specific PPG patterns rather than glucose-related changes
- ResNet34 is large, but provides the most accurate BGL estimation
- ResNet34 performed worse on the larger VitalDB test dataset compared to MUST dataset
  - Non-zero errors in Zone D/E
  - Expected since the paper acknowledges limited training data on hypo/hyperglycemia BGL

Non-invasive Glucose Monitoring

To-do:

- Investigate
  - Stronger regularization
  - Subject-independent validation
  - Data augmentation, especially for extreme BGLs
  - NIR wavelength selection
- Replicate research results using TensorFlow
  - Use VitalDB dataset
  - Downsample and preprocess PPG signals
  - Implement Algorithm 1 for Segmentation [1]
  - Run Inference and compare results with study

References

[1] M. Zeynali, K. Alipour, B. Tarvirdizadeh, and M. Ghamari, “Non-invasive blood glucose monitoring using PPG signals with various deep learning models and implementation using TinyML,” Scientific Reports, vol. 15, no. 581, 2025, doi: 10.1038/s41598-024-84265-8.